EZTime – Attendance & Payroll POC

Submission Document | Ido Shkuri

Live System: https://eztime-demo.onrender.com

Source Code: https://github.com/idoshkur/eztime-demo

Database: https://app.turso.tech/idoshkur/databases/eztime

Hosting: https://dashboard.render.com/web/srv-d6jcvqngi27c73d35le0

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AI Tools

This project was built with the follwoing AI tools:

- Claude Code - Used for the entire codebase implementation: backend, frontend, database, deployment, and debugging. All code was generated, reviewed, and iterated through Claude Code.
- ChatGPT -Used for initial PRD (Product Requirements Document) creation and product specification drafts.
- Gemini (Google) - Used for additional spec refinement and architectural brainstorming.

Part A: Architecture & Product Design (Theoretical)

Background & Core Challenges

Managing a workforce within a complex multi-site organization introduces two primary challenges:

1. Data Reliability: Preventing fraudulent reporting (e.g., remote check-ins) and ensuring verified employee identity.

2. Accurate Attribution: Ensuring working hours are assigned to the correct site and role, especially when employees perform multiple roles or work at different sites on the same day.

Selected Product Concept: Passive Reporter Model

The employee is defined as a passive reporter:

- The employee does **not** select: the site, the role, or the pay rate
- All shift parameters are predefined in the scheduling system by the employer
- The employee only confirms physical presence (check-in / check-out)

1. Technological Solutions for Data Collection & Reliability

Solution A: Physical Attendance Terminal (Biometric / Smart Card)

Description: A physical attendance terminal is installed at each work site, allowing employees to check in and out using fingerprint authentication or a smart employee card.

Operational Flow:

1. The manager defines a daily schedule in advance.
2. The employee arrives at the site and performs a clock-in at the terminal.
3. The server performs validation checks: an active schedule exists, entry time is within acceptable deviation limits.
4. The event is automatically linked to the predefined schedule.
5. The employee cannot modify the site or role.

Advantages	Disadvantages
Highest fraud prevention (prevents buddy punching and remote reporting)	Hardware costs (purchase, installation, maintenance)
No smartphone/battery/data plan required	Physical setup and deployment time
Simple "touch and go" UX	Less suitable for temporary sites
Works in GPS-restricted areas (underground, warehouses)	

RICE Evaluation:

- **Reach**: Very High (up to 100% of workforce)
- **Impact**: High (complete identity verification)
- **Confidence**: Very High (biometric/site-based validation)
- **Effort**: Medium (hardware logistics and installation)

Business Fit: Best for permanent, high-volume, or sensitive sites where maximum reliability is required.

Solution B: Location-Based Mobile Application

Description: A mobile application that allows attendance reporting only when the employee is within a predefined geographic radius of the assigned work site.

Operational Flow:

1. The manager defines a daily schedule.

2. The employee arrives and opens the app.

3. "Check-In" becomes active only if the employee is within the defined radius and an active schedule exists.

4. The event is stored with timestamp and location metadata.

Advantages	Disadvantages
Immediate deployment across thousands of sites	Privacy concerns (location tracking)
Minimal cost (software-only)	Requires smartphone with sufficient battery
Real-time push notifications	Moderate reliability (GPS can be inaccurate indoors)
Easily supports temporary sites	

RICE Evaluation:

- **Reach**: Medium–High (depends on workforce demographics)
- **Impact**: Medium–High (significant improvement over manual reporting)
- **Confidence**: Medium (GPS limitations)
- **Effort**: Low (software development only)

Business Fit: Best for temporary, distributed, or mobile workforce environments.

Recommended Business Approach

Given the sensitivity of payroll accuracy and the financial risk associated with incorrect role attribution, the chosen solution is Physical Attendance Terminals (Solution A).

This provides: the highest level of identity verification, elimination of personal device dependency, reliable on-site presence validation, and maximum payroll accuracy in role-based compensation environments.

2. Data Flow Architecture

Architectural Principle

The system follows an event-driven architecture and separates:

- **Raw Data Layer**: Attendance events as recorded
- **Business Processing Layer**: Hour calculation, overtime classification, and payroll preparation

This separation ensures scalability and flexibility for future policy changes.

Step 1: Event Capture (Client Layer)

The endpoint sends a raw event object (without hour calculations):

```
{
  "event_id": "uuid-12345",
  "employee_id": "E1001",
  "timestamp": "2026-02-04T07:15:00Z",
  "event_type": "CHECK_IN",
  "source": {
    "type": "BIOMETRIC_TERMINAL",
    "device_id": "TERM-SOUTH-04"
  },
  "verification": {
    "method": "FINGERPRINT",
    "status": "SUCCESS"
  }
}
```

Step 2: Validation & Alerting

If validation fails, an alert is generated:

```
{
  "alert_type": "UNAUTHORIZED_REPORT",
  "severity": "CRITICAL",
  "message": "Employee E1001 is not scheduled for SITE_SOUTH today",
  "actions": ["NOTIFY_MANAGER", "NOTIFY_EMPLOYEE_PUSH"],
  "original_event_id": "uuid-12345"
}
```

All events, including rejected ones, are stored for audit purposes.

Step 3: Work Session Builder

```
{
  "session_id": "sess-9988",
  "employee_id": "E1001",
  "site_id": "SITE_402",
  "role_id": "WAREHOUSE_OP",
  "actual_start": "2026-02-04T07:15:00Z",
  "actual_end": "2026-02-04T16:45:00Z"
}
```

Step 4: Daily Hours Classification

- Aggregate daily work time (split shifts combined)
- Determine overtime threshold:
- Default: overtime begins after 8 hours
- Night rule: if night_hours >= 2 (22:00–06:00), threshold drops to 7 hours
- Classify hours into tiers: 100% up to threshold, 125% to 10h, 150% above 10h

Step 5: Rate Assignment & Salary Calculation

Applied rate = MAX(hourly_rate) across all entries of the day.

Gross salary formula:

```
salary_100 = hours_100 × rate
salary_125 = hours_125 × rate × 1.25
salary_150 = hours_150 × rate × 1.5
gross_daily_salary = salary_100 + salary_125 + salary_150
```

Step 6: Audit & Transparency

All raw events are permanently stored, including rejected reports.

Step 7: Payroll-Ready Output

Final record includes totals, tiers, applied rate, gross salary, deficit, and breakdown by company and role.

Part B: Demo Implementation

Live System

URL: https://eztime-demo.onrender.com

The system is a fully functional POC for multi-company attendance & payroll calculation. It supports multiple entries per day, unique hourly rates per [employee + company + role], overtime tiers, night rules, deficit logic, and Excel import/export.

POC Note: There is currently **no login or role separation**. Any user can access all screens (including Admin) and perform all actions. In production, access would be controlled via authentication and RBAC.

Employee View

- Select employee
- Select date, company, role (from allowed lists)
- Enter start/end times (HH:MM)
- View daily payroll summary
- Export monthly payroll to Excel

Admin Panel

- Dashboard KPIs
- Upload Excel (upsert + dedupe)
- Manage Employees (CRUD + rates)
- Manage Time Entries (CRUD + validation)
- Data Insights (filters + breakdowns)
- Payroll Report (monthly + Excel export + export all employees)

Data Validation & Integrity

- Overlap detection (same-day + cross-day)
- Bulk upload overlap validation
- Atomic operations (transactions)
- Company/role authorization
- Rate verification
- Time format and duration validation
- Overnight shifts supported
- Duplicate prevention
- Consistent error handling

Bonus 1: API Design

Standard Error Format

```
{
  "error": {
    "code": "ERROR_CODE",
    "message": "Human-readable description of what went wrong",
    "details": {}
  }
}
```

Endpoint: Get Daily Payroll Analysis

Field	Value
URL	/api/payroll/daily
Method	GET
Purpose	Calculate and return a complete daily payroll analysis for a specific employee on a specific date.

Parameters

Parameter	Location	Required	Type	Description
employee_id	Query	Yes	string	Employee identifier
work_date	Query	Yes	string	Date in YYYY-MM-DD

Example Request

https://eztime-demo.onrender.com/api/payroll/daily?employee_id=E1001&work_date=2026-02-08

Success Response (200 OK)


```
{"employee_id":"E1001","work_date":"2026-02-08","standard_daily_quota":9,"total_worked_minutes":600,"total_hours":10,"night_minutes":0,"overtime_threshold":8,"hours_100":8,"hours_125":2,"hours_150":0,"applied_hourly_rate":68,"gross_daily_salary":714,"daily_deficit_hours":0,"entries":[{"id":"3e74d9fd-b454-40ea-a28c-c3a01ecb5e5f","work_date":"2026-02-08","employee_id":"E1001","company_name":"חברת בת ה","role_name":"מחסנאי","start_time":"08:00","end_time":"18:00","created_at":"2026-03-03T11:49:15.218Z"}],"breakdown_by_site_role":[{"company_name":"חברת בת ה","role_name":"מחסנאי","minutes":600,"hours":10,"entry_count":1}]}
```


Error Codes

Code	Status	Description
MISSING_PARAMS	400	Missing employee_id or work_date
EMPLOYEE_NOT_FOUND	404	Employee ID does not exist

Bonus 2: Full REST API (22 Endpoints)

Base URL: https://eztime-demo.onrender.com/api

#	Method	Endpoint	Purpose
1	GET	/api/employees	List all employees
2	POST	/api/employees	Create a new employee
3	GET	/api/employees/:id/options	Get allowed companies/roles
4	PUT	/api/employees/:id	Update employee details
5	DELETE	/api/employees/:id	Delete employee (cascade)
6	POST	/api/time-entries	Create time entry (validated)
7	GET	/api/time-entries?employee_id=...&work_date=...	Get day's entries
8	PUT	/api/time-entries/:id	Update time entry
9	DELETE	/api/time-entries/:id	Delete time entry
10	GET	/api/payroll/daily?employee_id=...&work_date=...	Daily payroll analysis
11	POST	/api/admin/upload	Upload Excel (multipart)
12	GET	/api/admin/dashboard	Dashboard KPIs
13	GET	/api/admin/employees	Employees with entry counts
14	GET	/api/admin/time-entries?page=...&limit=...	Paginated time entries
15	GET	/api/admin/rates?employee_id=...	List rates
16	POST	/api/admin/rates	Create rate
17	PUT	/api/admin/rates	Update rate
18	DELETE	/api/admin/rates	Delete rate
19	GET	/api/admin/insights?...	Filterable insights
20	GET	/api/admin/payroll-report?employee_id=...&month=...	Monthly payroll (JSON)
21	GET	/api/admin/payroll-report/export?employee_id=...&month=...	Monthly payroll (Excel)
22	GET	/api/admin/payroll-report/export-all?month=...	All employees payroll (Excel)

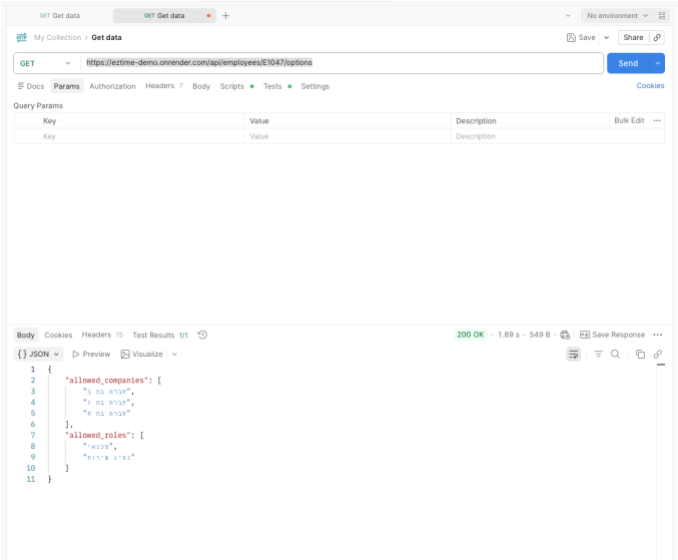
Error Codes Reference

Code	Status	Description
MISSING_FIELDS	400	Required fields not provided
MISSING_PARAMS	400	Required query params not provided
INVALID_DATE	400	Date not in YYYY-MM-DD format
INVALID_TIME	400	Time not in HH:MM format
INVALID_DURATION	400	Duration is 0 minutes
DURATION_TOO_LONG	400	Duration exceeds 16 hours
EMPLOYEE_NOT_FOUND	404	Employee ID doesn't exist
ENTRY_NOT_FOUND	404	Time entry UUID doesn't exist
RATE_NOT_FOUND	404	No rate for employee+company+role
COMPANY_NOT_ALLOWED	400	Employee not authorized for company
ROLE_NOT_ALLOWED	400	Employee not authorized for role
OVERLAP	400	Shift overlaps existing entry
EMPLOYEE_EXISTS	409	Employee ID already exists
RATE_EXISTS	409	Rate combo already exists
NO_FILE	400	No file uploaded
EXPORT_FAILED	500	Excel generation failed

Postman Screenshots

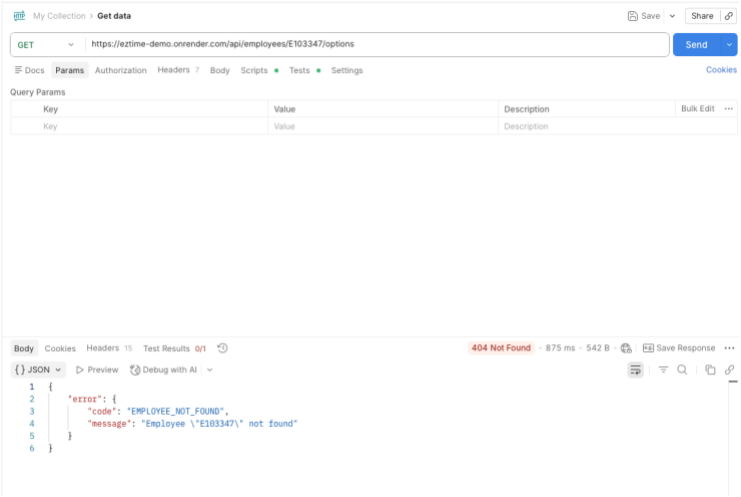
GET /api/employees/:id/options - Success (200 OK):

https://eztime-demo.onrender.com/api/employees/E1047/options



GET /api/employees/:id/options - Error (404 Not Found):

https://eztime-demo.onrender.com/api/employees/E103347/options



System Implementation Overview

Technology Stack

Layer	Technology	Reasoning
Backend	Node.js + Express + TypeScript	Fast development cycle, strong typing, large ecosystem
Frontend	React + Vite + TypeScript	Component-based UI, type safety, fast builds
Database	Turso (libSQL)	Cloud-hosted SQLite-compatible database, free tier, persistent
Deployment	Render.com	Free-tier web service with auto-deploy from GitHub
Excel I/O	xlsx (SheetJS)	Read uploaded .xlsx and generate payroll exports